7CS034 Lab 1 Visualization

1 Objectives

By the end of this session we will be able to

- Create attractive visuals for built in datasets in `R`.
- Understand some of the fundamentals of manipulating data in `R`.

2 Let's get started

R has several systems for making graphs, but `ggplot2` is one of the most elegant and most versatile. `ggplot2` implements the grammar of graphics, a coherent system for describing and building graphs. With `ggplot2`, you can do more and faster by learning one system and applying it in many places. This chapter will teach you how to visualize your data using ggplot2. We will start by creating a simple scatterplot and use that to introduce aesthetic mappings and geometric objects – the fundamental building blocks of ggplot2. We will then walk you through visualizing distributions of single variables as well as visualizing relationships between two or more variables. We'll finish off with saving your plots and troubleshooting tips. This section of the session is adapted directly from "R for Data Science". As usual, I recommedn that you read the relevant chapters to support your learning. If you feel comfortable, you can run ahead and work through the text at your own pace.

Let's start by installing `ggplot2` in case it is not installed already.

```
install.packages("tidyverse")
```

Then we load it

```
library(tidyverse)
```

You only need to install a package once, but you need to load it every time you start a new session. In addition to `tidyverse`, we will also use the `palmerpenguins` package, which includes the penguins dataset containing body measurements for penguins on three islands in the Palmer Archipelago, and the `ggthemes` package, which offers a colorblind safe color palette. You'll need to install and load both of these packages.

3 Penguins

Do penguins with longer flippers weigh more or less than penguins with shorter flippers? You probably already have an answer, but try to make your answer precise. What does the relationship between flipper length and body mass look like? Is it positive? Negative? Linear? Nonlinear? Does the relationship vary by the species of the penguin? How about by the island where the penguin lives? Let's create visualizations that we can use to answer these questions.

3.1 The penguins data frame

You can test your answers to those questions with the penguins data frame found in `palmerpenguins` . A data frame is a rectangular collection of variables (in the columns) and observations (in the rows). `penguins` contains 344 observations collected and made available by Dr. Kristen Gorman and the Palmer Station, Antarctica.
To make the discussion easier, let's define some terms:

- A variable is a quantity, quality, or property that you can measure.
- A value is the state of a variable when you measure it. The value of a variable may change from measurement to measurement.
- An observation is a set of measurements made under similar conditions (you usually make all of the measurements in an observation at the same time and on the same object).
- An observation will contain several values, each associated with a different variable. We'll sometimes refer to an observation as a data point.
- Tabular data is a set of values, each associated with a variable and an observation. Tabular data is tidy if each value is placed in its own "cell", each variable in its own column, and each observation in its own row.

In this context, a variable refers to an attribute of all the penguins, and an observation refers to all the attributes of a single penguin. Type the name of the data frame in the console and R will print a preview of its contents. Note that it says tibble on top of this preview. In the tidyverse, we use special data frames called tibbles that you will learn more about soon.

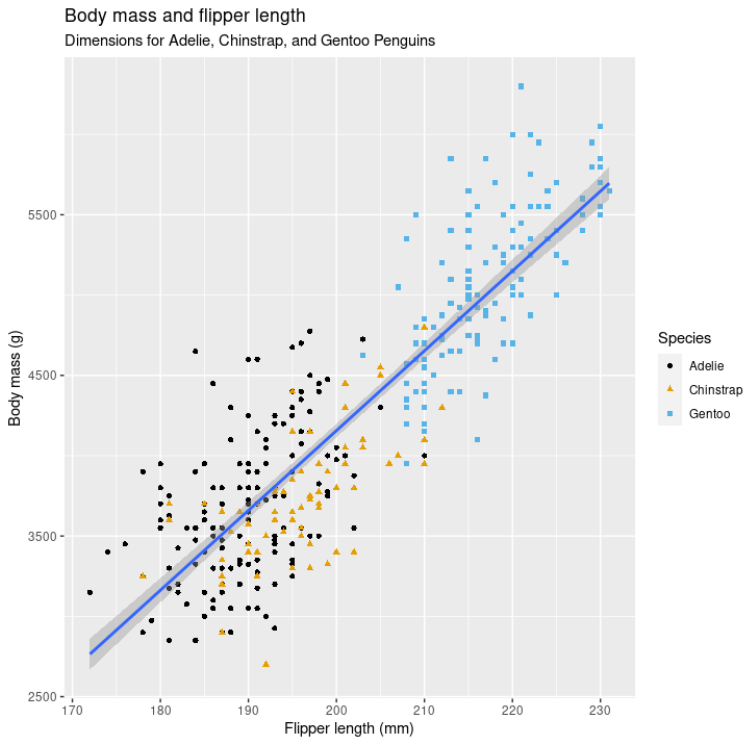
```
penguins
```

This data frame contains 8 columns. For an alternative view, where you can see all variables and the first few observations of each variable, run `View(penguins)` to open an interactive data viewer.
Among the variables in penguins are:

- `species`: a penguin's species (Adelie, Chinstrap, or Gentoo).
- `flipper_length_mm`: length of a penguin's flipper, in millimeters.

- `body_mass_g`: body mass of a penguin, in grams.

To learn more about penguins, open its help page by running `?penguins`. Our ultimate goal in this section is to recreate the following visualization displaying the relationship between flipper lengths and body masses of these penguins, taking into consideration the species of the penguin.



because this is our fuirst lab and we may never have used `R` before we will do thsi step by step. That way, you'll understand each of the commands we use.

3.2 Creating a ggplot

With `ggplot2`, you begin a plot with the function `ggplot()`, defining a plot object that you then add layers to. The first argument of `ggplot()` is the dataset to use in the graph and so `ggplot(data = penguins)` creates an empty graph that is primed to display the penguins data, but since we haven't told it how to visualize it yet, for now it's empty. This is not a very exciting plot, but you can think of it like an empty canvas you'll paint the remaining layers of your plot onto.

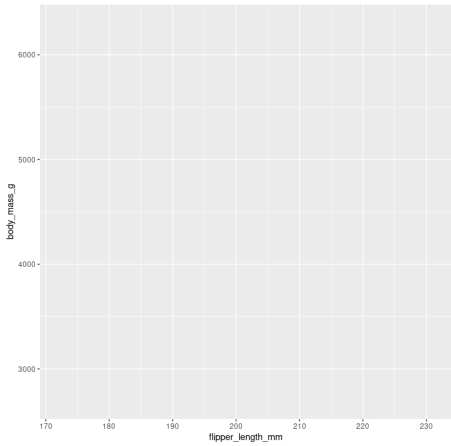
```
ggplot(data = penguins)
```

should produce

Next, we need to tell `ggplot()` how the information from our data will be visually represented. The mapping argument of the `ggplot()` function defines how variables in your dataset are mapped to visual properties (aesthetics) of your plot. The mapping argument is always defined in the `aes()` function, and the `x` and `y` arguments of `aes()` specify which variables to map to the `x` and `y` axes. For now, we will only map flipper length to the `x` aesthetic and body mass to the `y` aesthetic. `ggplot2` looks for the mapped variables in the data argument, in this case, `penguins`. The following plot shows the result of adding these mappings.

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
)
```

should produce



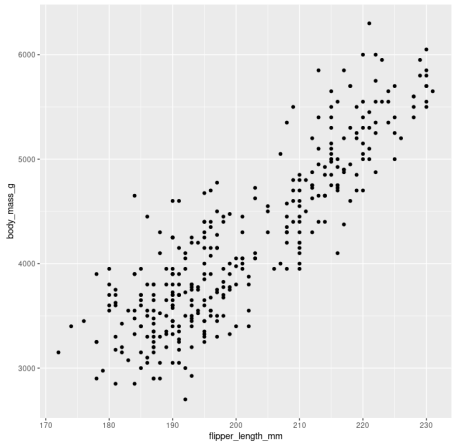
Our empty canvas now has more structure - it's clear where flipper lengths will be displayed (on the x-axis) and where body masses will be displayed (on the y-axis). But the penguins themselves are not yet on the plot. This is because we have not yet articulated, in our code, how to represent the observations from our data frame on our plot.

To do so, we need to define a **geom**: the geometrical object that a plot uses to represent data. These geometric objects are made available in `ggplot2` with functions

that start with `geom_`. People often describe plots by the type of geom that the plot uses. For example, bar charts use bar geoms (`geom_bar()`), line charts use line geoms (`geom_line()`), boxplots use boxplot geoms (`geom_boxplot()`), scatterplots use point geoms (`geom_point()`), and so on. The function `geom_point()` adds a layer of points to your plot, which creates a scatterplot. `ggplot2` comes with many geom functions that each adds a different type of layer to a plot. You'll learn a whole bunch of geoms by looking at Chapter 10 of "R for Data Science".

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
) +
  geom_point()
```

should produce



Now we have something that looks like what we might think of as a "scatterplot". It doesn't yet match our "ultimate goal" plot, but using this plot we can start answering the question that motivated our exploration: "What does the relationship between flipper length and body mass look like?" The relationship appears to be positive (as flipper length increases, so does body mass), fairly linear (the points are clustered around a line instead of a curve), and moderately strong (there isn't too much scatter around such a line). Penguins with longer flippers are generally larger in terms of their body mass. Before we add more layers to this plot, let's pause for a moment and review the warning message we got:

```
> ggplot(
+     data = penguins,
+     mapping = aes(x = flipper_length_mm, y = body_mass_g)
+ ) +
+     geom_point()
Warning message:
Removed 2 rows containing missing values (`geom_point()`).
>
```

We're seeing this message because there are two penguins in our dataset with missing body mass and/or flipper length values and `ggplot2` has no way of representing them on the plot without both of these values. `ggplot2` subscribes to the philosophy that missing values should never silently go missing. This type of warning is probably one of the most

common types of warnings you will see when working with real data – missing values are a very common issue. For the remaining plots in this section we will suppress this warning so it's not printed alongside every single plot we make.

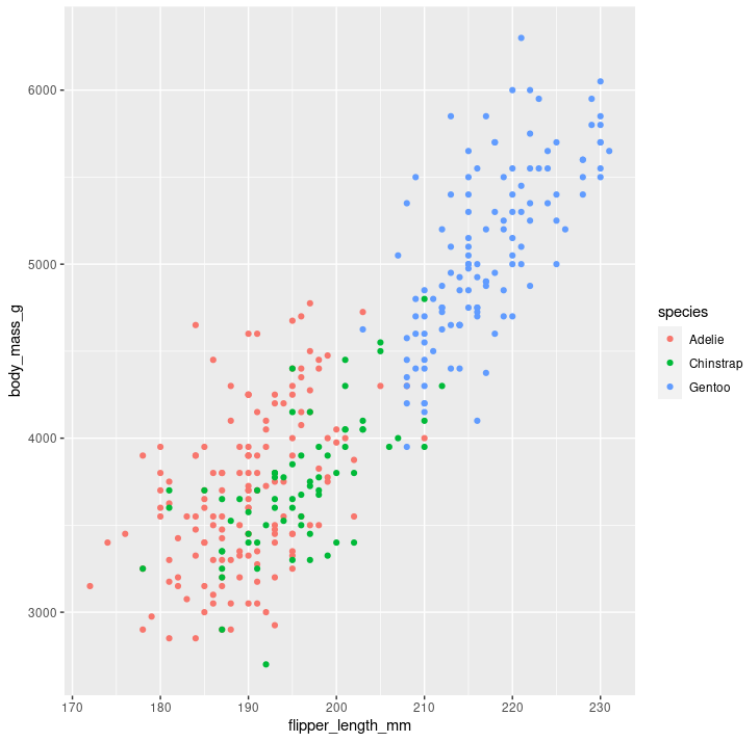
3.3 Adding aesthetics & layers

Scatterplots are useful for displaying the relationship between two numerical variables, but it's always a good idea to be skeptical of any apparent relationship between two variables and ask if there may be other variables that explain or change the nature of this apparent relationship. For example, does the relationship between flipper length and body mass differ by species? Let's incorporate species into our plot and see if this reveals any additional insights into the apparent relationship between these variables. We will do this by representing species with different colored points.

To achieve this, will we need to modify the aesthetic or the geom? If you guessed "in the aesthetic mapping, inside of `aes()`", you're already getting the hang of creating data visualizations with `ggplot2`! And if not, don't worry. Throughout the course you will make many more `ggplots` and have many more opportunities to check your intuition as you make them.

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g, color = species)
) +
  geom_point()
```

should produce



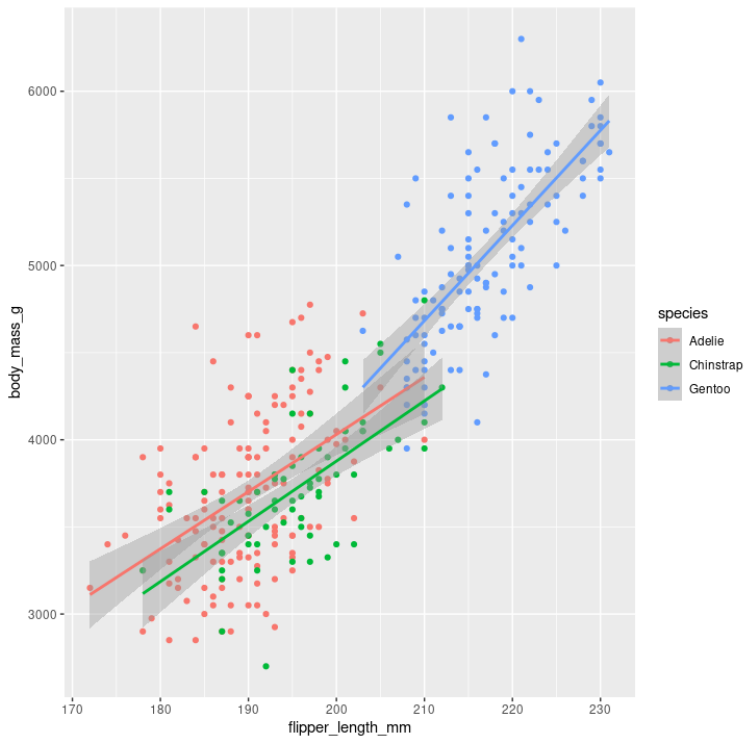
When a categorical variable is mapped to an aesthetic, `ggplot2` will automatically assign a unique value of the aesthetic (here a unique color) to each unique level of the variable (each of the three species), a process known as scaling. `ggplot2` will also add a legend that explains which values correspond to which levels.

Now let's add one more layer: a smooth curve displaying the relationship between body mass and flipper length. Before you proceed, refer back to the code above, and think about how we can add this to our existing plot.

Since this is a new geometric object representing our data, we will add a new geom as a layer on top of our point geom: `geom_smooth()`. And we will specify that we want to draw the line of best fit based on a linear model with `method = "lm"`.

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g, color = species)
) +
  geom_point() +
  geom_smooth(method = "lm")
```

should produce



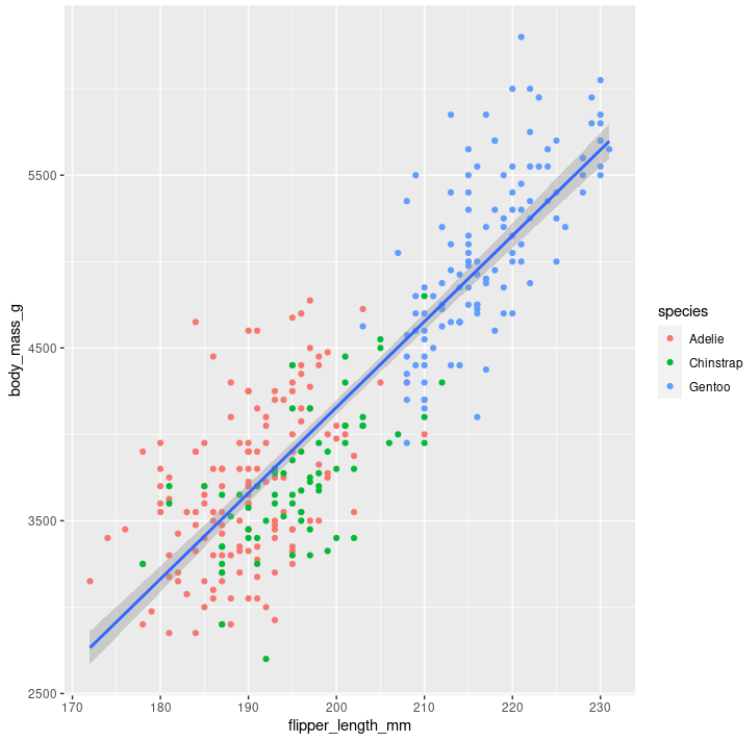
We have successfully added lines, but this plot doesn't look like the plot we want, which only has one line for the entire dataset as opposed to separate lines for each of the penguin species.

When aesthetic mappings are defined in `ggplot()`, at the global level, they're passed down to each of the subsequent geom layers of the plot. However, each geom function in `ggplot2` can also take a mapping argument, which allows for aesthetic mappings at the local level that are added to those inherited from the global level. Since we want points

to be colored based on species but don't want the lines to be separated out for them, we should specify `color = species` for `geom_point()` only.

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
) +
  geom_point(mapping = aes(color = species)) +
  geom_smooth(method = "lm")
```

should produce

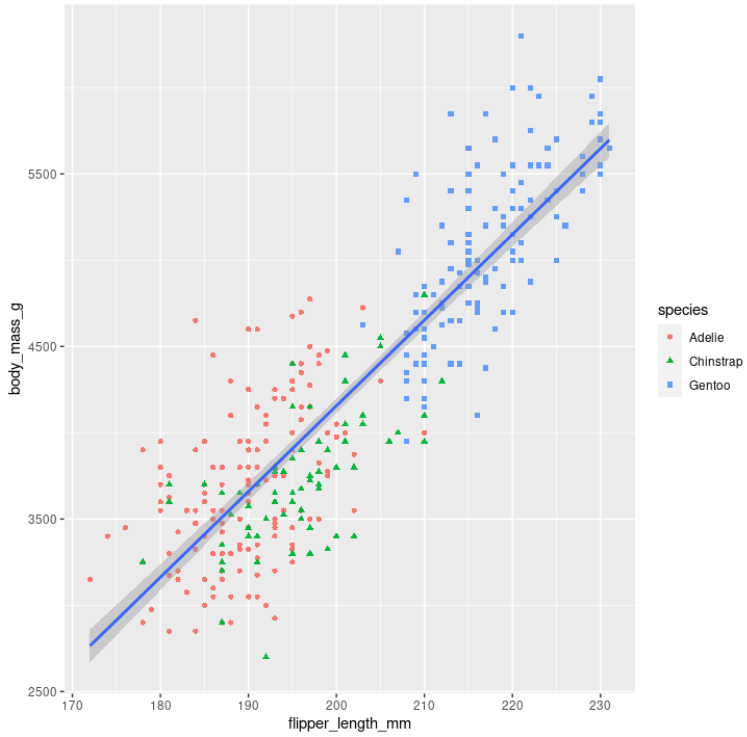


Voila! We have something that looks very much like our ultimate goal, though it's not yet perfect. We still need to use different shapes for each species of penguins and improve labels.

It's generally not a good idea to represent information using only colors on a plot, as people perceive colors differently due to color blindness or other color vision differences. Therefore, in addition to color, we can also map species to the shape aesthetic.

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
) +
  geom_point(mapping = aes(color = species, shape = species)) +
  geom_smooth(method = "lm")
```

should produce

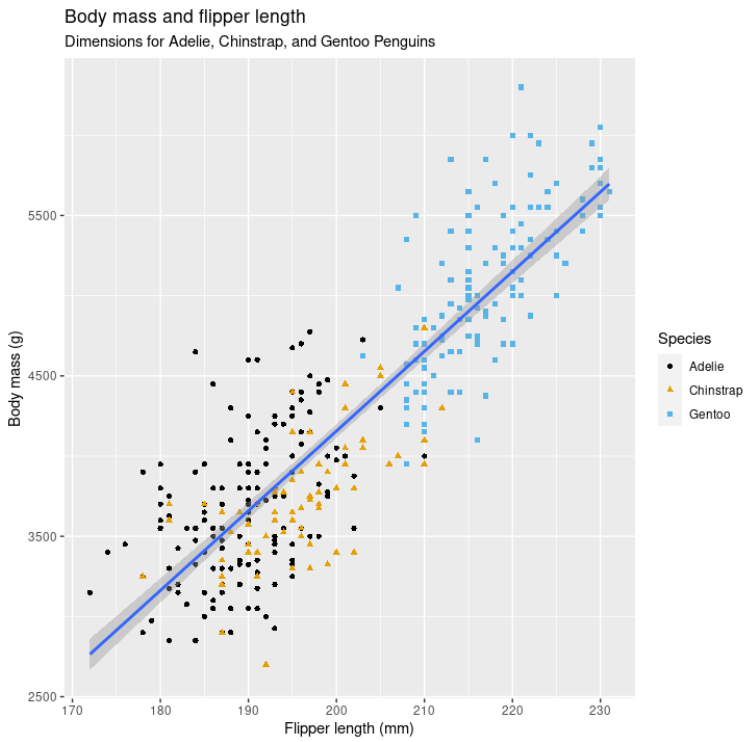


Note that the legend is automatically updated to reflect the different shapes of the points as well.

And finally, we can improve the labels of our plot using the `labs()` function in a new layer. Some of the arguments to `labs()` might be self explanatory: `title` adds a title and `subtitle` adds a subtitle to the plot. Other arguments match the aesthetic mappings, `x` is the x-axis label, `y` is the y-axis label, and `color` and `shape` define the label for the legend. In addition, we can improve the color palette to be colorblind safe with the `scale_color_colorblind()` function from the `ggthemes` package.

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
) +
  geom_point(mapping = aes(color = species, shape = species)) +
  geom_smooth(method = "lm")
```

should produce



We finally have a plot that perfectly matches our "ultimate goal"!

4 Exercises

Q1. How many rows are in penguins? How many columns? (Hint, use google to help you)

Q2. What does the `bill_depth_mm` variable in the penguins data frame describe? Read the help for `?penguins` to find out.

Q3. Make a scatterplot of `bill_depth_mm` vs. `bill_length_mm`. That is, make a scatterplot with `bill_depth_mm` on the y-axis and `bill_length_mm` on the x-axis. Describe the relationship between these two variables.

Q4. What happens if you make a scatterplot of `species` vs. `bill_depth_mm`? What might be a better choice of geom?

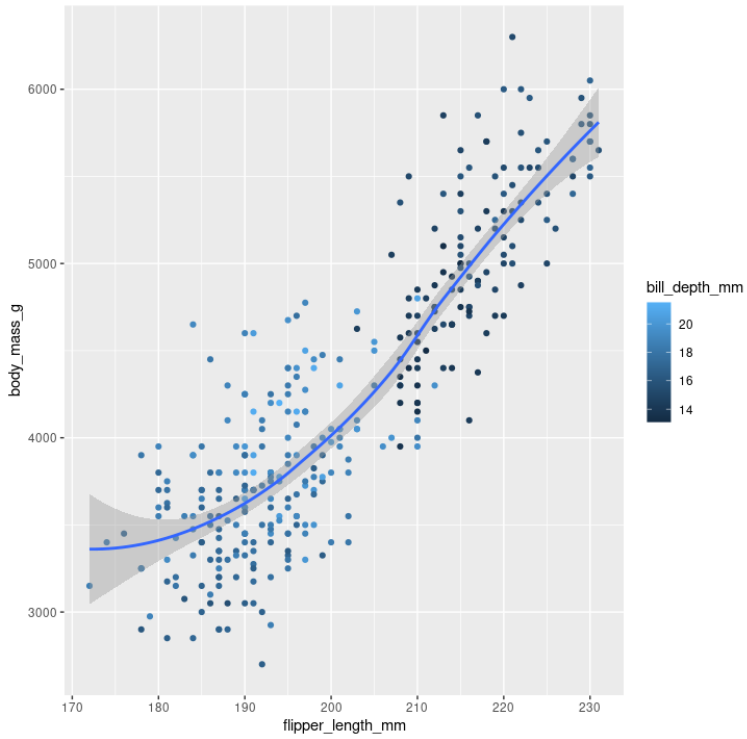
Q5. Why does the following give an error and how would you fix it?

```
> ggplot(data = penguins) +
+     geom_point()
Error in `geom_point()`:
! Problem while setting up geom.
i Error occurred in the 1st layer.
Caused by error in `compute_geom_1()`:
! `geom_point()` requires the following missing aesthetics: x and y
Run `rlang::last_trace()` to see where the error occurred.
>
```

Q6. What does the `na.rm` argument do in `geom_point()`? What is the default value of the argument? Create a scatterplot where you successfully use this argument set to `TRUE`.

Q7. Add the following caption to the plot you made in the previous exercise: "Data come from the `palmerpenguins` package." Hint: Take a look at the documentation for `labs()`.

Q8. Recreate the following visualization. What aesthetic should `bill_depth_mm` be mapped to? And should it be mapped at the global level or at the `geom` level?



Q9. Run this code in your head and predict what the output will look like. Then, run the code in `R` and check your predictions.

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g, color = island)
) +
  geom_point() +
  geom_smooth(se = FALSE)
```

Q10. Will these two graphs look different? Why/why not?

```
ggplot(
  data = penguins,
  mapping = aes(x = flipper_length_mm, y = body_mass_g)
```

```
) +
  geom_point() +
  geom_smooth()

ggplot() +
  geom_point(
    data = penguins,
    mapping = aes(x = flipper_length_mm, y = body_mass_g)
  ) +
  geom_smooth(
    data = penguins,
    mapping = aes(x = flipper_length_mm, y = body_mass_g)
  )
```

5 Over to you

Now it's your turn to replicate the `penguins` visualization process with the `iris` dataset and the `diamonds` dataset.